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SUBMITTED BY	ROLL NO
WAZMI ALI	2101103026
MOHD OWAIS KHAN	2201103903
SIDDHANT KUMAR SINGH	2101103020
PARAS MANDAL	2101103015

SCHOOL OF TECHNOLOGY
QUANTUM UNIVERSITY, ROORKEE



QUANTUM UNIVERSITY, ROORKEE

2024

DECLARATION

This is to certify that Software Engineering project report entitled "**Advanced Hospital Administration Health Care**" is the work carried out **Wazmi Ali, Mohd Owais Khan, Siddhant Kumar Singh, Paras Mandal.** students of **Diploma (Computer Science and Engineering)**, Submitted in the Department of Computer Science and Engineering at SCHOOL OF TECHNOLOGY, QUANTUM UNIVERSITY, University of Roorkee under the supervision of **DR. ASHISH SAINI, Department of Computer Science, QUANTUM UNIVERSITY.** This report has not been submitted to any other organization/institution for the award any other degree/diploma.

S.N.	NAME OF THE CANDIDATES	CANDIDATES SIGNATURE
1.	WAZMI ALI	-----
2.	MOHD OWAIS KHAN	-----
3.	SIDDHANT KUMAR SINGH	-----
4.	PARAS MANDAL	-----

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

PARUL TYAGI
(Project Guide)

ACKNOWLEDGEMENT

Even though it is said in our Vedas, wake up and do not stop till you achieve your goal. But still, no ordinary person can complete his journey on the struggle path of life alone. Whenever a person starts his journey, he meets many people on the way to the destination, who walks with him, encourages him, helps him, and supports him. The success of my research work, apart from my personal efforts, is largely based on the encouragement of many of my well-wishers and the inspiration and guidance shown by them. First of all, I would like to express my gratitude for the supernatural power to which we name God, by whose grace I have been able to write this.

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1. Wazmi Ali
2. Mohd Owais Khan
3. Siddhant Kumar Singh
4. Paras Mandal

ABSTRACT

Our project, the **Hospital Management System**, encompasses a comprehensive approach to hospital operations, including patient registration, data storage, and appointment scheduling with doctors. The software assigns a unique ID to every patient, ensuring that all patient and staff details are systematically stored and easily accessible. Users can search for doctor availability and retrieve patient information using the assigned IDs. Access to the system is controlled through a secure login requiring a username and password, ensuring that only authorized administrators or receptionists can input data into the database. The system is designed with a user-friendly interface that facilitates quick and easy data retrieval, while also ensuring that personal data remains secure and protected.

The software comprises two main modules: the Administration Level and the User Level, which includes both patients and doctors. The application employs stringent authentication protocols to control access. Administrative tasks involve managing doctor and patient information, supported by a database designed specifically for these purposes. User complaints are addressed by the appropriate authorities, ensuring a responsive support system. The Patient Module allows users to check appointments, view prescriptions, and pay doctors' fees online, streamlining various hospital processes and enhancing overall efficiency.

Table of Content

Sno.	Title	Page no.
	DECLARATION	2
	ACKNOWLEDGEMENT	3
	ABSTRACT	4
	LIST OF FIGERS	7
CHAPTER 1	INTRODUCTION	8 to 16
1.1	Problem Introduction	8
1.2	Literature Survey	9
1.3	Working	9
1.4	Purpose	12
1.5	Scope	13
1.6	Overview	15
CHAPTER 2	LITERATURES REVIEW	18 to 26
2.1	Existing Systems	18
2.2	Methodology & Source code	19-24
CHAPTER 3	SPECIFIC REQUIREMENTS	27 to 30
3.1	Performance Requirements	27
3.2	Views	28
3.3	Viewpoints	28
3.4	Perspective	29
3.5	Implementation	29
CHAPTER 4	SAMPLE SCREENSHORT	31 to 37

4.1	Home Page	31
4.2	Doctors	32
4.3	Services	33
4.4	Appointment Book	34
4.5	Message	35
4.6	Feedback for patient	36
4.7	Cocomo Model	37
CHAPTER 5	RISK ANALYSIS	38 to 38
CHAPTER 6	TESTING	39 to 44
6.1	Type of Testing	39
6.2	Maintenance and Support	40
6.3	Security	42
CHAPTER 7	CONCLUSION	45 to 46
7.1	Reference	46

LIST OF FIGURES

Sno.	Title	Page no.
2.1	Architectural Diagram	20
2.2	Use case Diagram	21
2.3	Apache Server Monitor	25
4.1	Home Page	31
4.2	Doctors Page	32
4.3	Services	33
4.4	Appointment Book	34
4.5	Doctor List	34
4.6	Drop us a Message	35
4.7	Feedback Form of Patient	36

CHAPTER 1

INTRODUCTION

The project Hospital Management system includes registration of patients, storing their details into the system, and also computerized billing in the pharmacy, and labs. The software has the facility to give a unique id for every patient and stores the details of every patient and the staff automatically. It includes a search facility to know the current status of each room. User can search availability of a doctor and the details of a patient using the id. The Hospital Management System can be entered using a username and password. It is accessible either by an administrator or receptionist. Only they can add data into the database. The data can be retrieved easily. The interface is very user-friendly. The data are well protected for personal use and makes the data processing very fast. Hospital Management System is powerful, flexible, and easy to use and is designed and developed to deliver real conceivable benefits to hospitals. Hospital Management System is designed for multispecialty hospitals, to cover a wide range of hospital administration and management processes. It is an integrated end-to-end Hospital Management System that provides relevant information across the hospital to support effective decision making for patient care, hospital administration and critical financial accounting, in a seamless flow. Hospital Management System is a software product suite designed to improve the quality and management of hospital management in the areas of clinical process analysis and activity-based costing. Hospital Management System enables you to develop your organization and improve its effectiveness and quality of work. Managing the key processes efficiently is critical to the success of the hospital helps you manage your processes.

1.1 Problem Introduction:

Lack of immediate retrievals the information is very difficult to retrieve and to find particular information like- E.g. - To find out about the patient's history, the user has to go through various registers. This results in in convenience and wastage of time.

Lack of immediate information storage the information generated by various transactions takes time and efforts to be stored at right place.

Lack of prompt updating various changes to information like patient details or immunization details of child are difficult to make as paper work is involved.

Error prone manual calculation Manual calculations are error prone and take a lot of time this may result in incorrect information. For example, calculation of patient's bill based on various treatments. Preparation of accurate and prompt reports: - This becomes a difficult task as information is difficult to collect from various register.

1.2 Literature Survey

The literature survey delves into a diverse array of existing hospital management systems, offering insights into their strengths, weaknesses, and areas ripe for improvement. Across various studies, a consensus emerges regarding the pivotal role of technology adoption in shaping the landscape of hospital management. While electronic health records (EHR) systems have facilitated advancements in data accessibility and clinical decision-making, challenges persist in terms of interoperability, usability, and data security.

- ❖ Patient registration.
- ❖ Appointment Scheduling.
- ❖ Billing and Invoicing.
- ❖ Electronic Medical records.
- ❖ Inventory management.
- ❖ Reporting and Analytics.
- ❖ Pharmacy management.

Moreover, the literature sheds light on the critical link between technology adoption and patient satisfaction levels. Studies consistently demonstrate that innovative technologies, such as telemedicine platforms and patient portals, can enhance patient engagement, communication, and convenience. However, disparities in digital literacy, access to technology, and cultural factors underscore the importance of designing inclusive solutions that cater to diverse patient populations.

In addition to technology-driven initiatives, the literature underscores the importance of optimizing operational efficiency in hospital management. Lean management principles, rooted in the elimination of waste and continuous improvement, have garnered significant attention for their potential to streamline workflows, reduce costs, and enhance resource utilization. Furthermore, studies emphasize the need for robust performance measurement frameworks to track key indicators such as patient wait times, staff productivity, and clinical outcomes, enabling data-driven decision-making and accountability.

Furthermore, the literature survey identifies several emerging trends and areas of focus in hospital management research. These include the integration of artificial intelligence and machine learning algorithms for predictive analytics, the implementation of value-based care models to incentivize quality over quantity, and the exploration of innovative care delivery models such as accountable care organizations and patient-centered medical homes.

1.3 Working

A Hospital Management System (HMS) is a complex software solution designed to manage the various aspects of hospital operations. The system typically consists of

multiple integrated modules that handle different functions, from patient registration to billing and inventory management. Here's a detailed explanation of how the HMS works:

1.3.1. Patient Registration and Admission

❖ Patient Registration:

- When a patient arrives at the hospital, their personal information is collected and entered into the system.
- Details such as name, age, gender, contact information, and insurance details are recorded.
- A unique patient ID is generated for future reference.

❖ Admission Process:

- If the patient needs to be admitted, the HMS facilitates the allocation of a bed, room, and other necessary resources.
- Admission details, including the reason for admission, attending physician, and expected length of stay, are recorded.

1.3.2. Appointment Scheduling

❖ Appointment Booking:

- Patients can schedule appointments through the hospital's website, patient portal, or directly at the hospital.
- The system displays available time slots for different doctors and departments.
- Appointments are confirmed and reminders are sent to patients via SMS or email.

❖ Doctor's Schedule Management:

- The HMS helps manage the schedules of doctors, ensuring that appointments are allocated efficiently.
- It prevents overbooking and ensures optimal utilization of the doctor's time.

1.3.3. Electronic Medical Records

❖ Record Maintenance:

- The HMS maintains comprehensive electronic medical records for each patient.
- These records include medical history, diagnosis, treatment plans, test results, and progress notes.

❖ Access and Updates:

- Authorized healthcare providers can access and update patient records in real-time.
- The system ensures that the latest information is always available, improving the quality of care.

1.3.4. Billing and Claims Processing

- **Billing:**
- The HMS automates the billing process, generating accurate bills based on services rendered.
- It includes charges for consultations, tests, procedures, medications, and other services.

❖ Insurance Claims:

- The system facilitates the submission of insurance claims.
- It verifies patient insurance details and ensures that claims are processed efficiently.
- Automated follow-ups on pending claims are also managed by the system.

1.3.5. Laboratory and Diagnostic Management

❖ Test Ordering and Scheduling:

- Doctors can order lab tests and diagnostic procedures directly through the HMS.
- The system schedules the tests and informs the relevant departments.

❖ Results Management:

- Test results are entered into the system as soon as they are available.
- Doctors and patients can access these results through their respective portals.

1.3.6. Pharmacy and Inventory Management

❖ Pharmacy Operations:

- The HMS manages the hospital pharmacy, including prescription handling and medication dispensing.
- It checks for drug interactions and ensures that prescriptions are filled accurately.

❖ Inventory Management:

- The system tracks inventory levels of medications, medical supplies, and equipment.

- It generates alerts for low-stock items and facilitates reordering to ensure that the hospital is always well-stocked.

1.3.7. Reporting and Analytics

❖ Data Analytics:

- The HMS analyses data to generate insights into hospital performance, patient outcomes, and operational efficiency.
- It identifies trends and patterns that can help in decision-making and strategic planning.
- **Reporting:**
- The system generates various reports, such as financial reports, patient demographic reports, and operational efficiency reports.
- These reports can be customized to meet the specific needs of hospital management and regulatory requirements.

1.3.8. Security

❖ Data Security:

- The HMS employs robust security measures to protect patient data, including encryption, access controls, and audit trails.
- It ensures that only authorized personnel have access to sensitive information.

1.3.9. Patient Portal

❖ Patient Access:

- The patient portal allows patients to access their medical records, view test results, and schedule appointments online.
- Patients can also communicate with their healthcare providers and receive educational materials.

❖ Feedback and Engagement:

- The portal provides a platform for patients to give feedback on their experience, helping the hospital improve its services.
- It also enhances patient engagement by providing them with easy access to their health information and care team.

1.4 Purpose

The purpose of this project is to design, develop, and implement a comprehensive Hospital Management System (HMS) that significantly enhances the operational efficiency, patient care, and overall management of a healthcare facility. This purpose is driven by several key motivations:

- ❖ **Operational Efficiency:** Hospitals face complex administrative and clinical processes that require significant time and resources to manage. The purpose of the HMS is to streamline these processes through automation and integration, reducing the administrative burden on healthcare staff and allowing them to focus more on patient care.
- ❖ **Improved Patient Care:** Accurate and accessible patient information is crucial for delivering high-quality healthcare. The HMS aims to enhance patient care by providing healthcare providers with comprehensive and real-time access to patient records, thereby improving clinical decision-making and patient outcomes.
- ❖ **Financial Management:** Efficient financial management is vital for the sustainability of any healthcare facility. The HMS aims to optimize billing processes, reduce errors in financial transactions, and improve the overall revenue cycle management. This ensures that hospitals can operate smoothly without financial constraints impacting patient care.
- ❖ **Data Security and Compliance:** With the increasing amount of sensitive patient information being stored electronically, data security and regulatory compliance have become critical concerns. The purpose of the HMS is to implement robust security measures that protect patient data and ensure compliance with healthcare regulations such as the Health Insurance Portability and Accountability Act (HIPAA).
- ❖ **Integration and Interoperability:** Modern healthcare requires seamless integration and interoperability between various systems and departments within a hospital. The HMS aims to provide a platform that can easily integrate with other healthcare systems and third-party applications, promoting efficient data exchange and collaboration across different healthcare providers and institutions.
- ❖ **Resource Management:** Efficient utilization of hospital resources such as staff, equipment, and facilities is essential for optimal hospital operation. The purpose of the HMS is to provide tools for effective resource management, ensuring that resources are used efficiently and are readily available when needed.
- ❖ **Regulatory Compliance:** Hospitals must comply with numerous local, national, and international regulations. The HMS is designed to facilitate compliance with these regulations by providing necessary documentation, tracking, and reporting capabilities.
- ❖ **Future-Proofing Healthcare Operations:** The healthcare industry is continuously evolving with advancements in technology and changes in regulatory frameworks. The purpose of this HMS is to provide a scalable and adaptable solution that can evolve with these changes, ensuring long-term sustainability and relevance.

1.5 Scope

The scope of a Hospital Management System (HMS) covers a wide range of functionalities and modules designed to manage various aspects of hospital operations. This comprehensive scope ensures that all critical areas of hospital management are addressed, leading to improved efficiency, better patient care, and streamlined operations. Here are the key components of the scope of an HMS:

1.5.1. Patient Management

- ❖ **Patient Registration:** Captures detailed patient information and assigns unique patient IDs for easy tracking and retrieval.
- ❖ **Appointment Scheduling:** Manages appointment bookings, rescheduling, and cancellations, including automated reminders for patients.
- ❖ **Admission and Discharge:** Handles the processes for admitting patients, assigning beds, and managing discharge procedures along with discharge summaries.

1.5.2. Medical Records Management

- ❖ **Electronic Medical Records (EMR):** Maintains comprehensive electronic medical records that include medical history, diagnoses, treatment plans, and progress notes.
- ❖ **Clinical Documentation:** Facilitates the creation, storage, and management of clinical notes, charts, and other relevant medical documentation.
- ❖ **Lab Results and Imaging:** Stores and integrates laboratory test results, radiology images, and other diagnostic data into patient records for easy access and review.

1.5.3. Pharmacy Management

- ❖ **Medication Management:** Oversees the prescription and dispensing of medications, ensuring accuracy and compliance with medical guidelines.
- ❖ **Inventory Control:** Tracks medication inventory levels, manages procurement, and monitors usage to prevent shortages and overstock situations.
- ❖ **Drug Interaction Alerts:** Provides alerts for potential drug interactions, allergies, and contraindications to ensure patient safety.

1.5.4. Laboratory and Diagnostic Management

- ❖ **Test Ordering and Scheduling:** Facilitates the ordering and scheduling of laboratory tests and diagnostic procedures.
- ❖ **Results Management:** Records and manages test results, making them easily accessible to healthcare providers and patients.
- ❖ **Equipment Management:** Tracks and maintains diagnostic equipment, ensuring regular calibration and servicing.

1.5.5. Operational Management

- ❖ **Resource Scheduling:** Manages the schedules of healthcare providers, operating rooms, and other hospital resources to optimize utilization.
- ❖ **Staff Management:** Handles staff rosters, attendance, and workload distribution, ensuring efficient staff deployment.
- ❖ **Facility Management:** Oversees the maintenance and management of hospital facilities, including housekeeping, utilities, and infrastructure.

1.5.6. Reporting and Analytics

- ❖ **Clinical Reporting:** Generates reports on clinical activities, patient outcomes, and treatment effectiveness.
- ❖ **Operational Analytics:** Analyzes operational data to identify trends, optimize workflows, and improve overall efficiency.
- ❖ **Regulatory Reporting:** Ensures compliance with regulatory requirements by producing necessary reports and documentation for audits and inspections.

1.5.7. Patient and Staff Engagement

- ❖ **Patient Portal:** Provides patients with access to their medical records, appointment scheduling, and communication with healthcare providers.
- ❖ **Feedback Mechanisms:** Collects and manages patient feedback to improve service quality and patient satisfaction.
- ❖ **Staff Training and Development:** Offers training modules and resources to support continuous staff development and adherence to best practices.

1.5.8. Security and Compliance

- ❖ **Data Security:** Implements robust security measures, including encryption, access controls, and audit trails, to protect patient data.
- ❖ **Regulatory Compliance:** Ensures the system adheres to healthcare regulations and standards, such as HIPAA, for data handling and privacy.
- ❖ **Audit Trails:** Maintains detailed records of all transactions and changes to provide a clear audit trail, simplifying regulatory reporting and audits.

1.5.9. Integration and Interoperability

- ❖ **System Integration:** Integrates with other healthcare systems, such as laboratory information systems, radiology systems, and electronic health record (EHR) systems.
- ❖ **Interoperability:** Ensures seamless data exchange with external healthcare providers and organizations to facilitate coordinated care.
- ❖ **API Support:** Provides APIs for third-party application integration, enhancing system functionality and flexibility.

1.5.10. Support for Research and Development

- ❖ **Data Analytics for Research:** Utilizes data analytics to support clinical research, identify trends, and improve healthcare practices and outcomes.
- ❖ **Clinical Trials Management:** Manages and monitors clinical trials, ensuring compliance with protocols and regulatory requirements.

1.6 Overview

The healthcare industry is one of the most important sectors of our economy. It is responsible for providing care to hundreds and billions of people around the world. The health of all people is paramount to their well-being, which means that the healthcare industry is an important part of society as a whole.

Technology has played an increasingly more important role in the delivery of healthcare. **Hospital management system** has emerged to be an evolving healthcare technology trend in recent times. A HMS is a comprehensive solution for healthcare facilities to gather patient data, secure vital records, automate scheduling, and more!

Advantages of a Hospital Management System

- ❖ **Operational Efficiency:** Automates administrative tasks, reducing paperwork and streamlining workflows.
- ❖ **Enhanced Patient Care:** Provides accurate and real-time access to patient records, improving clinical decision-making.
- ❖ **Financial Management:** Automates billing and claims processing, ensuring timely and accurate financial transactions.
- ❖ **Data Security:** Implements robust security measures to protect patient information and ensure compliance with regulations.
- ❖ **Resource Management:** Optimizes the utilization of hospital resources, such as staff, equipment, and facilities.
- ❖ **Integration and Interoperability:** Facilitates seamless data exchange and integration with other healthcare systems and third-party applications.
- ❖ **Improved Reporting and Analytics:** Generates detailed clinical and operational reports, aiding in better decision-making and regulatory compliance.
- ❖ **Patient Satisfaction:** Enhances patient experience by reducing wait times and improving the overall quality of care.

Disadvantages of a Hospital Management System

- ❖ **High Initial Cost:** Significant investment required for purchasing and implementing the system.
- ❖ **Complex Implementation:** The deployment process can be complex and time-consuming, requiring extensive planning and resources.
- ❖ **Training Requirements:** Staff need to be trained to use the new system, which can be time-consuming and costly.
- ❖ **Technical Issues:** Potential for technical problems, such as software bugs or system downtime, which can disrupt hospital operations.
- ❖ **Data Privacy Concerns:** Risk of data breaches and unauthorized access to sensitive patient information.
- ❖ **Maintenance Costs:** Ongoing costs associated with system maintenance, updates, and technical support.

CHAPTER 2

LITERATURES REVIEW

Hospital Management Systems are critical technological solutions that streamline hospital operations, improve patient care, and enhance overall healthcare delivery. The literature on HMS encompasses various studies that analyse their implementation, benefits, challenges, and impact on healthcare facilities. This review synthesizes key findings from recent literature to provide a comprehensive understanding of the development, adoption, and efficacy of HMS.

2.1 Existing Systems

An existing Hospital Management System provides insights into the development, capabilities, benefits, and challenges of current systems. Understanding these aspects helps identify gaps and opportunities for improvement in designing and implementing a new HMS.

2.1.1 Key Components

- ❖ **Patient Management:** Handles patient registration, admission, discharge, and transfer processes.
- ❖ **Electronic Medical Records (EMR):** Maintains comprehensive patient medical histories, including diagnoses, treatments, and lab results.
- ❖ **Billing and Financial Management:** Manages patient billing, insurance claims, and financial reporting.
- ❖ **Pharmacy Management:** Oversees medication prescribing, dispensing, and inventory control.
- ❖ **Laboratory Management:** Automates lab test orders, results reporting, and specimen tracking.
- ❖ **Appointment Scheduling:** Facilitates appointment booking and resource allocation.

2.1.2 Benefits

- ❖ **Improved Efficiency:** Automation of administrative and clinical processes reduces manual workload, minimizes errors, and accelerates operations.
- ❖ **Enhanced Patient Care:** Access to accurate, real-time patient data supports better clinical decision-making and continuity of care.
- ❖ **Financial Management:** Streamlined billing and insurance processes improve cash flow and reduce revenue cycle times.
- ❖ **Data Security:** Advanced encryption and access control mechanisms protect patient data and ensure compliance with regulations.
- ❖ **Interoperability:** Integration with other healthcare systems and third-party applications promotes seamless data exchange and collaboration.

2.1.3 Challenges and Limitations

- ❖ **High Costs:** The initial investment and ongoing maintenance costs for HMS can be substantial.
- ❖ **Complex Implementation:** Deploying an HMS involves significant planning, customization, and training, which can be time-consuming and resource-intensive.
- ❖ **User Adoption:** Resistance to change among healthcare staff can hinder the successful implementation of new systems.
- ❖ **Technical Issues:** System downtime, software bugs, and integration problems can disrupt hospital operations.
- ❖ **Data Privacy Concerns:** Ensuring the security of sensitive patient information remains a critical challenge, especially with increasing cyber threats.

2.2 METHODOLOGY

2.2.1 Introduction

Systems are designed keeping in mind an issue that is to be solved. Every system is designed in its unique keeping in mind the requirement of the problem or the issue. Our system solves the problem of searching for the good that the customer's needs. System design involves the design of overall architecture, based on which we design components, modules and interfaces. The beginning of any system architecture is by decomposing it into smaller fragments. Decomposition and binding of components makes the architecture easy to understand and makes it easier to understand. Our system uses algorithm for collecting data which will collect the data of user and we have data analysing algorithm which will analyse and highlight the needs of user.

2.2.2 ARCHITECTURE DIAGRAM

An architectural diagram is a diagram of a system that is used to abstract the overall outline of the software system and the relationships, constraints, and boundaries between components. It is an important tool as it provides an overall view of the physical deployment of the software system and its evolution roadmap.

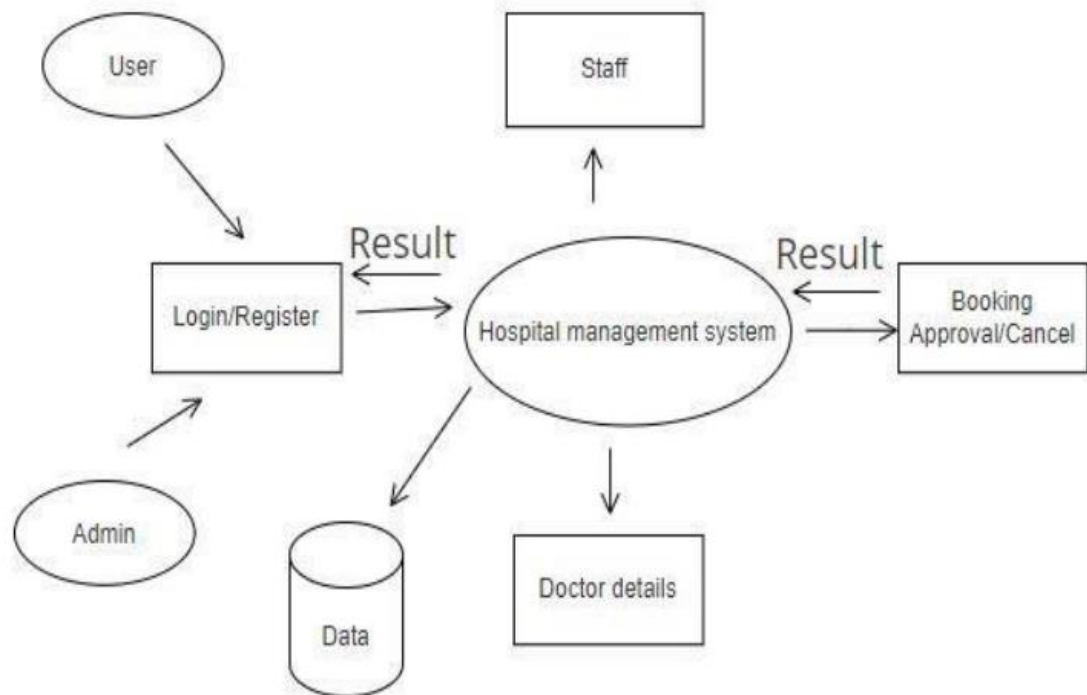


Fig 2.1: Architectural Diagram

2.2.3 Use Case Diagram

A use case diagram in the Unified Modelling Language (UML) is a type of behavioural diagram defined by and created from a use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases.

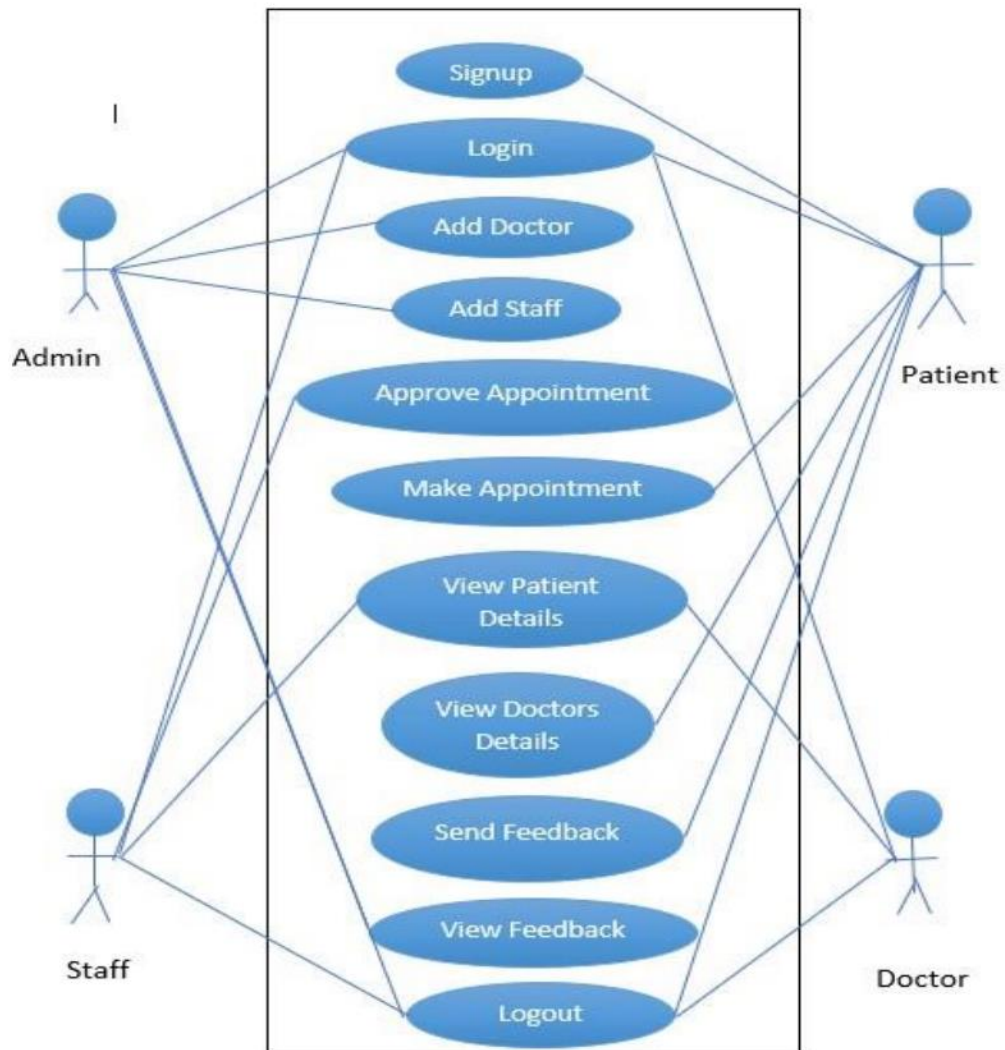


Fig 2.2: Use Case Diagram

2.2.4 Introduction to Tools used in Project

2.2.5 Introduction of Html

Hyper-Text Mark-up Language (HTML) is a simple mark-up system used to create hypertext documents that are portable from one platform to another. 19 HTML documents are SGML documents with generic semantics that are appropriate for representing information from a wide range of applications. HTML mark-up can represent hypertext news, mail, documentation, and hypermedia; menus of options; database query results; simple structured documents with in-lined graphics; and hypertext views of existing bodies of information. **Advantages of Html**

- 1 It is widely used.
- 2 Every browser supports HTML language.
- 3 Easy to learn and use.

4 It is by default in every window so we don't need to purchase extra software.

2.2.6 Introduction of CSS

Cascading Style Sheets, fondly referred to as CSS, is a simply designed language intended to simplify the process of making web pages presentable. CSS allows you to apply styles to web pages. More importantly, CSS enables you to do this independent of the HTML that makes up each web page.

Advantages of CSS

1. Greater consistency in design.
2. Ease of presenting different styles to different viewers.

2.2.7 About PHP:

PHP: Hypertext Pre-processor is a widely used, general-purpose scripting language that was originally designed for web development to produce dynamic web pages. For this purpose, PHP code is embedded into the HTML source code and can be interpreted but a web server with PHP processor module, which helps to generate web page document. PHP is a general-purpose scripting language that is especially suited to server-side web development where PHP generally runs on a web server. It can also be used for command line scripting and client-side GUI application. Many operating system and platforms, can be used with many relational database management systems. It is also free of charge.

Working with PHP:

When a client requests web page containing PHP code from the server, then the requested PHP pages are passed under PHP environment and interaction with database is made if required. After server-side processing, the resulting HTML pages are passed to client and displayed on the browser. In this way the working of PHP is complete.

2.2.8 Introduction to MySQL:

MySQL is a Relational Database Management System (RDBMS) that runs as a server providing multi-user access to a number of databases. MySQL is pronounced (MY S-Q-L)

MySQL development project has made its source available under the terms of General Public License. MySQL is owned and sponsored by a single for profit firm, the Swedish company MySQL AB, now owned by Sun Microsystems, a subsidiary of Oracle.

Corporation. MySQL works on many different system platforms including AIX, BSD i, FreeBSD, HP/UX, i5/OS, Linux, Mac OS X, Net BSD, Novell NetWare, Open BSD, Open Solaris, e com Station , OS/2 Wrap, QNX, IRIX, Solaris, Symbian, SunOS, SCO Open Server, SCO Unix Ware, Sano's, Tru64 and Microsoft Windows. A port of MySQL to Open VMS also exists. All major programming languages with language-specific APIs include Libraries for accessing MySQL database.

2.2.9 SOURCE CODE

```
<?php

$name = $_POST['name'];
$phone = $_POST['phone'];
$email = $_POST['email'];
$age = $_POST['age'];
$address = $_POST['address'];
$hidden = date("Y-m-d");
$message = $_POST['message'];
$option = $_POST['MyDoctor'];
$hostname = "localhost";
$username = "root";
$password = "";
$databse = "unity_healthcare";
$conn = new mysqli($hostname, $username, $password, $databse);
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "INSERT INTO book_appointment (name, phone, email, age, address, date,
message, Doctor)

VALUES ('$name', '$phone', '$email', '$age', '$address', '$hidden',
'$message','$option')";
if ($conn->query($sql) === TRUE) {
    echo "Appointment book successfully";
}
else {
    echo "Error Try Again: " . $sql . "<br>" . $conn->error;
}

$conn->close();
?>
```

```

<?php
$con=mysqli_connect("localhost","root","","unity_healthcare");
if(isset($_POST['btnSubmit']))
{
    $name = $_POST['txtName'];
    $email = $_POST['txtEmail'];
    $contact = $_POST['txtPhone'];
    $message = $_POST['txtMsg'];

    $query="insert into contact(name,email,contact,message)
values('$name','$email','$contact','$message');";
    $result = mysqli_query($con,$query);

    if($result)
    {
        echo '<script type="text/javascript">';
            echo 'alert("Message sent successfully!");';
            echo 'window.location.href = "Emergency.html"';
            echo '</script>';
        }
    }
?>

```


2.2.10 Introduction to APACHE SERVER

In this project Apache server is user to parse and execute PHP pages, before deploying websites on the server, the website should be tested at the developer's side to get a feel of how the website will work on actual server. Therefore, Apache server is like a local server on the developer side, Apache server should be informed about the environment on which it should work. In our project Apache server is configured to work with PHP, in this way all the PHP pages are parsed and executed by the server.

- ❖ To used platform like Visual Studio Code, XAMPP, and phpMyAdmin database, you will need to have a good understanding of PHP, HTML, and CSS. You will also need to be familiar with the XAMPP server and how to use it to create and manage databases.



Fig 2.3: Apache Server Monitor

2.2.11 Expected Outcome

The implementation of the outlined strategies is anticipated to yield a multitude of tangible and intangible benefits, ultimately resulting in a transformative impact on hospital management and patient care delivery. Foremost among these expected outcomes is the enhancement of patient care quality, achieved through more informed clinical decision-making, personalized treatment plans, and streamlined care processes. By centralizing patient data through electronic health records (EHR) systems and leveraging data analytics tools, healthcare providers can deliver more proactive, evidence-based care that is tailored to individual patient needs.

- ❖ Additionally, the adoption of these strategies is poised to alleviate administrative burdens and optimize resource allocation within hospital settings. Through the automation of routine tasks, streamlined workflows, and the application of lean management principles, hospitals can reduce inefficiencies, minimize waste, and reallocate resources towards frontline patient care initiatives. This, in turn, enhances overall operational efficiency and ensures that healthcare resources are utilized judiciously to meet patient needs effectively.
- ❖ Furthermore, the anticipated outcomes include improved communication between healthcare providers, facilitated by the integration of telemedicine solutions and collaborative platforms. Enhanced communication channels enable seamless coordination of care, timely exchange of information, and interdisciplinary collaboration, ultimately leading to more comprehensive and integrated patient care delivery.
- ❖ Moreover, the system aims to elevate patient satisfaction levels by improving accessibility to healthcare services, enhancing communication with healthcare providers, and delivering a more patient-centric care experience. By leveraging technology-enabled solutions and implementing patient engagement strategies, hospitals can empower patients to take an active role in their healthcare journey, thereby fostering greater satisfaction and loyalty.

CHAPTER 3

SPECIFIC REQUIREMENTS

3.1 PERFORMANCE REQUIREMENTS

- ❖ **Response time-** The system will give responses within 1 second after checking the patient information and other information.
- ❖ **Capacity-** The system must support 1000 people at a time
- ❖ **User interface-** User interface screen will response within 5 seconds

SAFETY REQUIREMENTS

If there is extensive damage to a wide portion of the database due to catastrophic failure, such as a disk crash, the recovery method restores a past copy of the database that was backed up to archival storage and reconstructs a more current state by reapplying or redoing the operations of committed transactions from the backed up log, up to the time of failure. All the administrative and data entry operators have unique logins so system can understand who is login in to system right now no intruders allowed except system administrative nobody cannot change record and valuable data.

3.1.2. SECURITY REQUIREMENTS

- ❖ Want take the responsibility of failures due to hardware malfunctioning.
- ❖ Warranty period of maintaining the software would be one year.
- ❖ Additional payments will be analysed and charged for further maintenance.
- ❖ If any error occur due to a user's improper use. Warranty will not be allocated to it.
- ❖ No money back returns for the software.

3.1.3 SOFTWARE SYSTEM ATTRIBUTES

- ❖ **Usability:** Software can be used again and again without distortion.
- ❖ **Availability:** The system shall be available all the time.
- ❖ **Correctness:** Bug free software which fulfils the correct need/requirements of the client.
- ❖ **Maintainability:** The ability to maintain, modify information and update fix problems of the system.
- ❖ **Accessibility:** Administrator and many other users can access the system but the access level is controlled for each user according to their work scope.

3.2 Views

3.2.1. Functional View This view defines the functional requirements of the HMS. It specifies the tasks that the system should be able to perform, the roles of different users, and the interactions between users and the system. The functional view documents the features and capabilities of the system, such as patient registration, appointment scheduling, billing, and medical records management.

3.2.2. Technical View This view describes the technical aspects of the HMS. It includes the system architecture, hardware and software requirements, and the integration of the system with other hospital systems. The technical view documents the technical specifications of the system and provides guidelines for system installation, configuration, and maintenance.

3.2.3. Process View This view focuses on the processes involved in the HMS. It describes the workflows, procedures, and policies that govern the operation of the system. The process views document the steps involved in each process, including the inputs, outputs, and the roles of the users involved.

3.2.4. User View This view presents the system from the perspective of the users. It includes user manuals, training materials, and user interface design guidelines. The user view documents the user interface, the interactions between the users and the system, and the user experience.

3.2.5. Data View This view defines the data structures and relationships within the HMS. It includes the database schema, data dictionaries, and data flow diagrams. The data view documents the data elements, their attributes, and their relationships.

3.2. 6. Security View This view addresses the security aspects of the HMS. It includes security policies, access control mechanisms, and encryption techniques. The security view documents the security features of the system and provides guidelines for ensuring the confidentiality, integrity, and availability of the data.

3.3 Viewpoints

3.3.1. Functional Viewpoint This viewpoint focuses on the system's functional requirements and represents the system's functionality from the perspective of hospital staff and patients. Some views that can be included in this viewpoint are patient management, appointment scheduling, billing, prescription management, inventory management, and medical records management.

3.3.2. Technical Viewpoint This viewpoint focuses on the technical infrastructure that supports the hospital management system. It includes views of the hardware, software, and network architecture that make up the system, as well as the protocols and standards used for data communication and storage.

3.3.3. Process Viewpoint This viewpoint focuses on the processes that are used to manage and operate the hospital management system. It includes views of the

workflows, procedures, and policies that govern the use of the system, as well as the roles and responsibilities of the various users and stakeholders.

3.3.4. User Viewpoint This viewpoint focuses on the needs and requirements of the users of the hospital management system. It includes views of the user interface design, user experience, and usability testing, as well as user training and support.

3.3.5. Data Viewpoint This viewpoint focuses on the data that is managed and processed by the hospital management system. It includes views of the data models, data dictionaries, and data flow diagrams, as well as data security and privacy policies.

3.3.6. Security Viewpoint This viewpoint focuses on the security and privacy of the hospital management system. It includes views of the security policies, procedures, and protocols that are used to protect the system from unauthorized access, data breaches, and other security threats.

3.4 Perspective

3.4.1. Functional Perspective This perspective outlines the functional requirements of the HMS, including the tasks the system should be able to perform, roles of different users, and interactions between users and the system. It provides documentation for system features and capabilities such as patient registration, appointment scheduling, billing, and medical records management.

3.4.2. Technical Perspective The technical perspective outlines the technical aspects of the HMS, including system architecture, hardware and software requirements, and system integration with other hospital systems. This perspective provides documentation for the technical specifications of the system and provides guidelines for installation, configuration, and maintenance.

3.4.3. Process Perspective The process perspective outlines the workflows, procedures, and policies that govern the operation of the HMS. It provides documentation for the steps involved in each process, including inputs, outputs, and user roles.

3.4.4. User Perspective the user perspective focuses on the system from the perspective of the users, including user manuals, training materials, and user interface design guidelines. It provides documentation for the user interface, user-system interactions, and user experience.

3.4.5. Data Perspective The data perspective defines the data structures and relationships within the HMS, including database schema, data dictionaries, and data flow diagrams. It provides documentation for data elements, attributes, and their relationships.

3.4.6. Security Perspective The security perspective focuses on the security aspects of the HMS, including security policies, access control mechanisms, and encryption

techniques. It provides documentation for security features and guidelines for ensuring data confidentiality, integrity, and availability.

3.5 Implementation

3.5.1. Functional Viewpoint To implement the functional perspective to the functional viewpoint, identify the functional requirements of the system from the perspective of the hospital staff and patients. Then design views, such as use case diagrams, sequence diagrams, and activity diagrams, to represent those functional requirements.

3.5.2. Technical Viewpoint To implement the technical perspective to the technical viewpoint, identify the hardware, software, and network infrastructure needed to support the hospital management system. Then design views, such as system architecture diagrams and network topology diagrams, to represent the technical infrastructure.

3.5.3. Process Viewpoint To implement the process perspective to the process viewpoint, identify the workflows, procedures, and policies that govern the use of the hospital management system. Then design views, such as process flow diagrams, swim lane diagrams, and state machine diagrams, to represent those processes.

3.5.4. User Viewpoint To implement the user perspective to the user viewpoint, identify the needs and requirements of the different types of users who will interact with the hospital management system, such as hospital staff and patients. Then design views, such as wireframes, user interface mock-ups, and user flow diagrams, to represent the user interface and user experience.

3.5.5. Data Viewpoint To implement the data perspective to the data viewpoint, identify the types of data that will be managed and processed by the hospital management system, such as patient records, medical histories, and billing information. Then design views, such as ER diagrams, data flow diagrams, and data dictionaries, to represent the data models and data flows.

3.5.6. Security Viewpoint To implement the security perspective to the security viewpoint, identify the security and privacy policies, procedures, and protocols that are needed to protect the hospital management system from unauthorized access, data breaches, and other security threats. Then design views, such as security architecture diagrams, threat models, and access control matrices, to represent the security infrastructure and protocols.

CHAPTER 4

SAMPLE SCREENSHOTS

4.1 Home Page



Fig 4.1: Home Page

4.2 Doctors

TopDoctors



Dr. Sandeep Vaishya
Specialist: Spine & Neurosurgeon

HOD & Executive Director Sandeep Vaishya is a renowned Neurosurgeon in India with more than 22 years of experience in the field having worked with some of the top institutions and hospitals of India.



Dr. Mohamed Rela
Specialist: Liver Transplant Surgeon & HPB Surgeon

Chairman & Managing Director, Dr. Rela Institute & Medical Centre, Chennai Prof. Mohamed Rela is a pioneer in liver transplantation with over 28+ years of experience, In December 1997.



Dr. Subhash Gupta
Specialist: Liver Transplant Surgeon

Subhash Gupta is Chairman of Max Centre for Liver and Biliary Sciences which is one of the largest Liver Transplant Centres in the world with experience of over 3000 transplants.

Doctors



Dr. Naresh Trehan
Speciality: Cardiology, Cardiac Surgeon

The Chairman and Managing Director of Medanta – The Medicity, Dr. Naresh Trehan is a world-renowned cardiovascular and cardiothoracic surgeon. He has been awarded the highly prestigious Padma Bhushan.



Dr. Hari Goyal
Specialist: Medical Oncologist

Hari Goyal had the privilege to work in the best of Cancer Centers of India. He started his career in the field of Oncology at Tata Memorial Centre & did the prestigious course of DM from All India Institute of Medical Sciences.



Dr. H.S. Chhabra
Specialist: Spine Surgeon

Dr. H S Chhabra is one of the coveted & most experienced surgeons in the field of Spine Surgery in India. He has over 27+ years of experience and has performed over 3700 spinal surgeries MBBS.

Fig 4.2: Doctors Page

4.3 Services

Unity offers many services and wellness programs to help support and provide the tools patients need to hit your personal health and behavioural health goals. Speak with your provider about whether these services or programs are a good fit for you and/or your family.

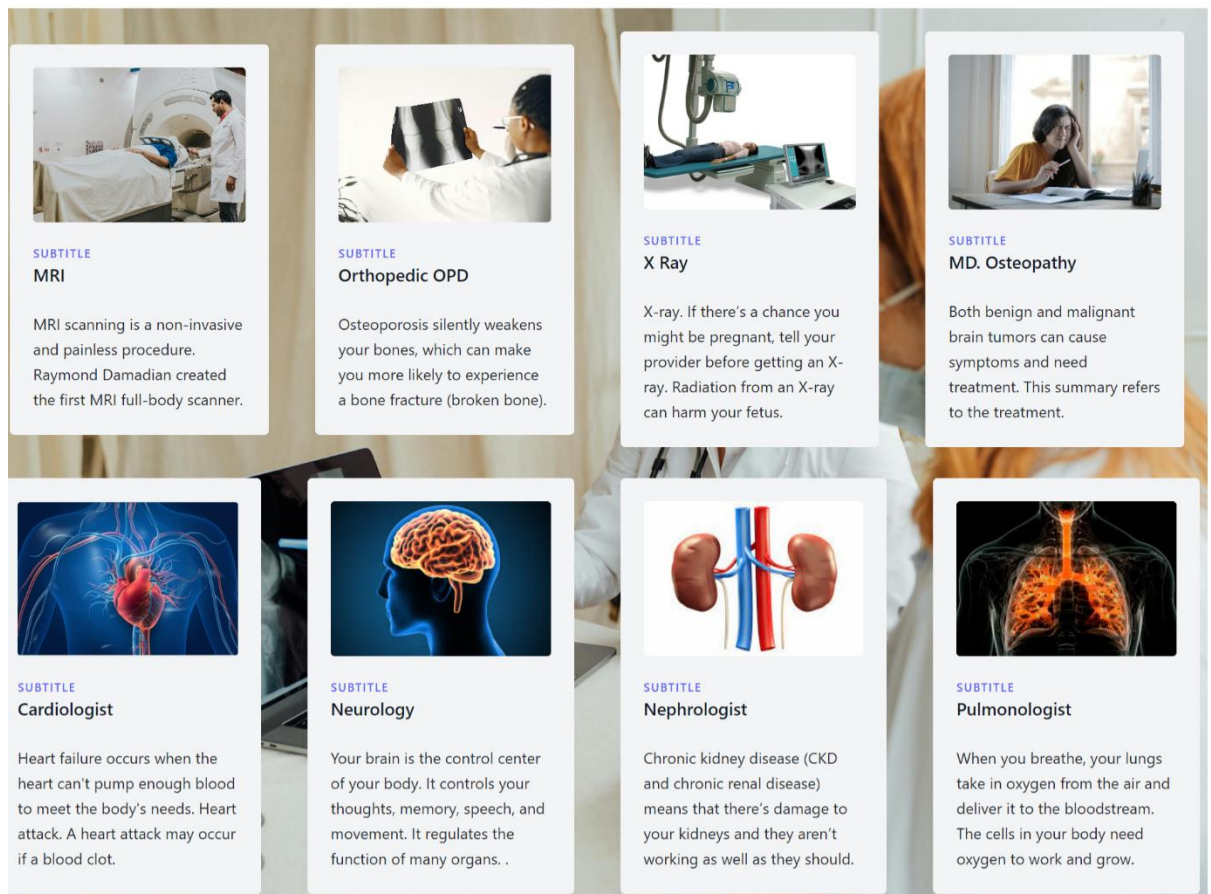
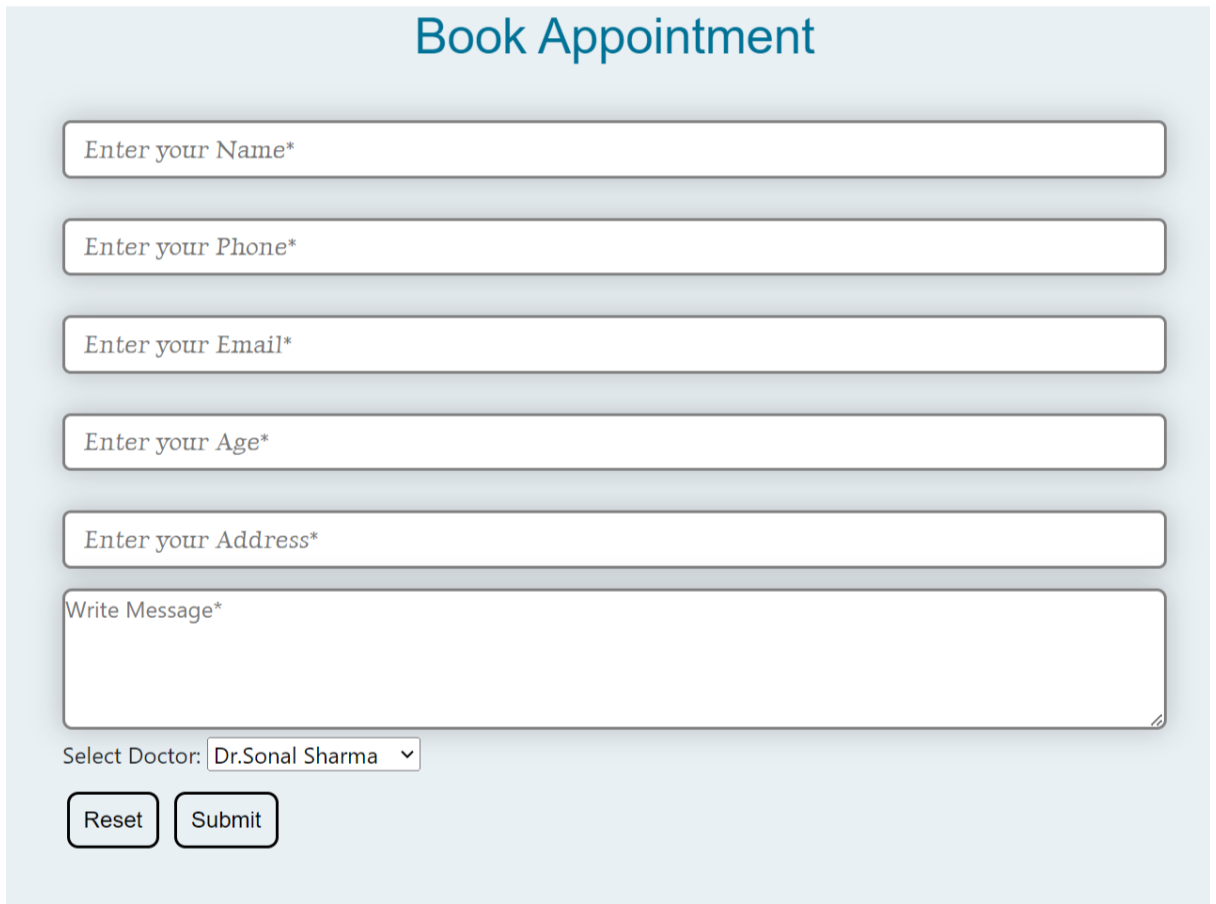


Fig 4.3: Services

4.4 Appointment Booking



The image shows a web form titled "Book Appointment" in a blue header. The form contains several input fields: "Enter your Name*", "Enter your Phone*", "Enter your Email*", "Enter your Age*", and "Enter your Address*", all with light blue borders. Below these is a larger text area labeled "Write Message*" with a small cursor icon at the bottom right. At the bottom left, there is a "Select Doctor:" label followed by a dropdown menu showing "Dr.Sonal Sharma" with a downward arrow. To the right of the dropdown are two buttons: "Reset" and "Submit", both with rounded corners and light blue borders.

Fig 4.4: Appointment Booking

Select Doctor:

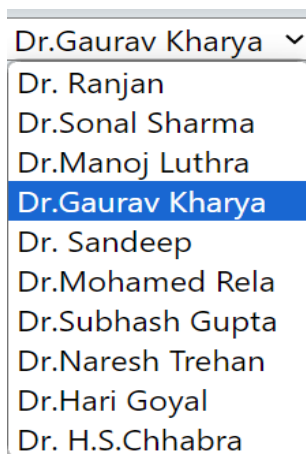
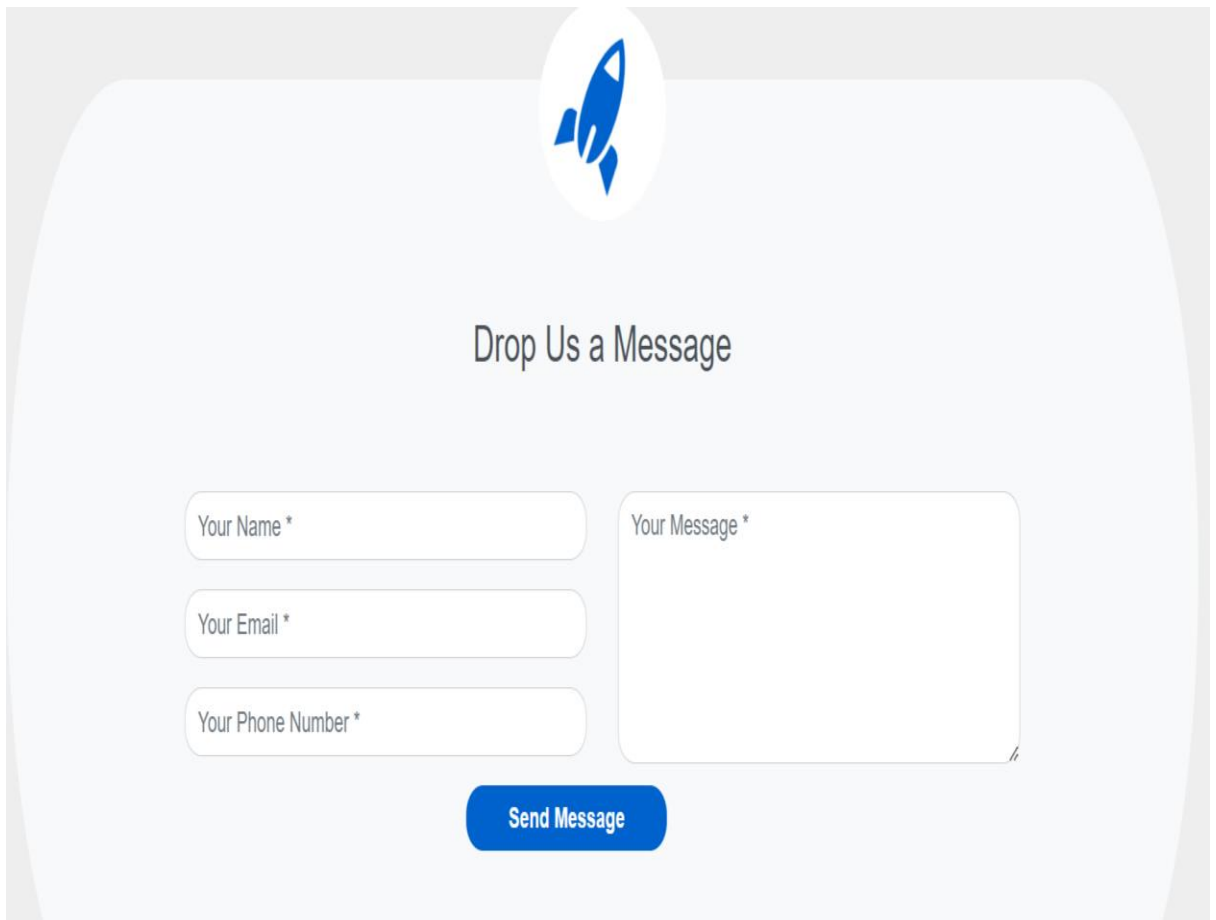
- 
- A dropdown menu for selecting a doctor. The menu is open, showing a list of doctor names. The top item, "Dr.Gaurav Kharya", is highlighted with a blue background and a white downward arrow. The other names are in black text on a white background.
- Dr.Gaurav Kharya ▾
 - Dr. Ranjan
 - Dr.Sonal Sharma
 - Dr.Manoj Luthra
 - Dr.Gaurav Kharya
 - Dr. Sandeep
 - Dr.Mohamed Rela
 - Dr.Subhash Gupta
 - Dr.Naresh Trehan
 - Dr.Hari Goyal
 - Dr. H.S.Chhabra

Fig 4.5: Doctor List

4.5 Message

The image shows a web form titled "Drop Us a Message". At the top center is a blue rocket icon inside a white circle. Below the title, there are three input fields on the left: "Your Name *", "Your Email *", and "Your Phone Number *". To the right of these is a larger text area labeled "Your Message *". At the bottom center is a blue button with the text "Send Message".

Drop Us a Message

Your Name *

Your Email *

Your Phone Number *

Your Message *

Send Message

Fig 4.6: Drop us a Message

Drop Us a Message

We're here to help with any questions or compliant you may have. Whether you're looking for more information about our services or need assistance, feel free to reach out. Our team is dedicated to providing you with prompt and efficient support. Use the form below to send us a message, and we'll get back to you as soon as possible. Thank you for connecting with us!

4.6 Feedback Form for patient

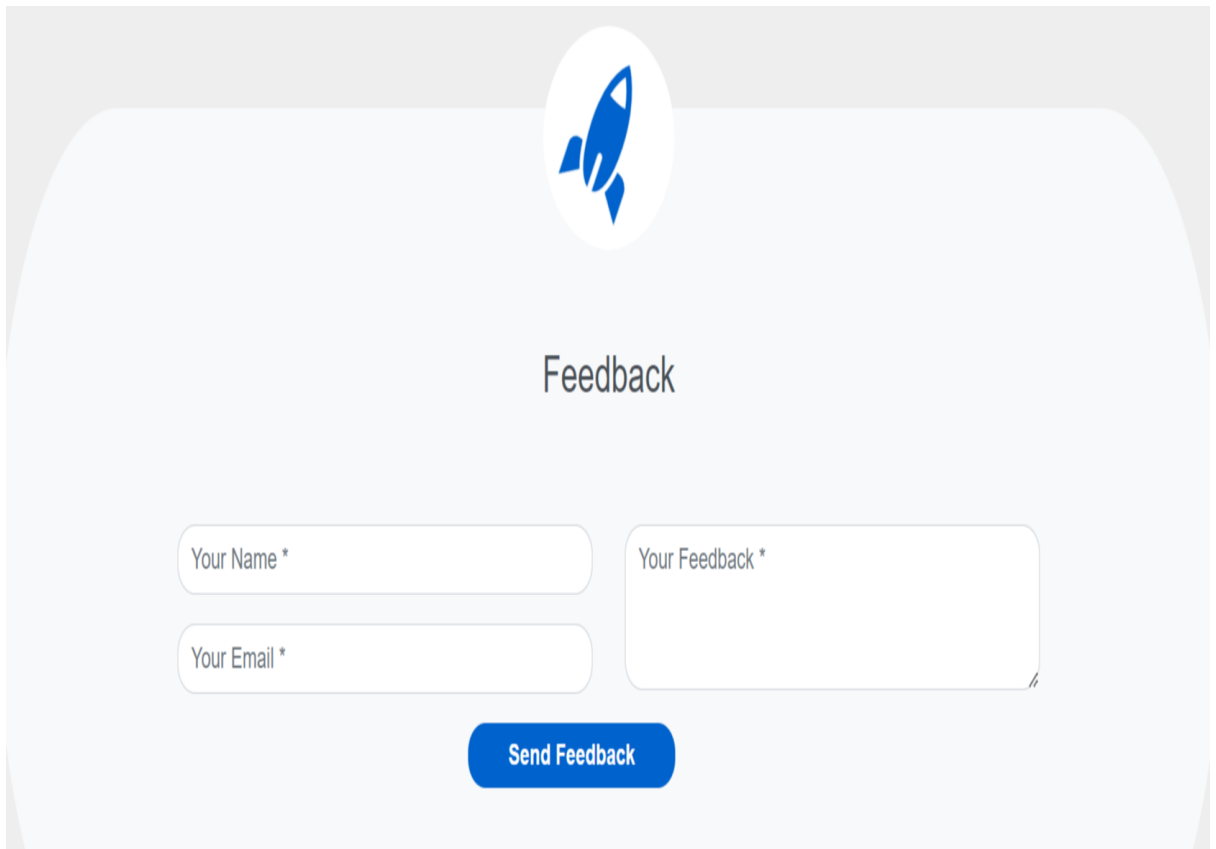
The image shows a web-based feedback form. At the top center is a blue rocket icon inside a white circle. Below the icon, the word "Feedback" is centered in a large, dark grey font. Underneath, there are three input fields: "Your Name *" (a single-line text box), "Your Email *" (a single-line text box), and "Your Feedback *" (a larger multi-line text area). All three fields have a light grey border and a small asterisk indicating they are required. Below these fields is a blue button with the text "Send Feedback" in white.

Fig 4.7: Feedback Form for patient

Patient Feedback

Patient Feedback I recently had the opportunity to receive care at Unity Health Care, and I am extremely impressed with the high level of service provided. The hospital staff were not only professional and knowledgeable but also displayed genuine compassion and kindness. The facilities were clean and well-maintained, ensuring a comfortable and safe environment for recovery. I was particularly pleased with the efficient and seamless processes, from admission to discharge. Thank you, Unity Health Care Hospital, for your excellent care and commitment to patient well-being.

4.7 COCOMO MODEL

The original COCOMO model became one of the most widely used and discussed software cost estimation models in the industry. It has evolved into a more comprehensive estimation model, called COCOMO II. Models require sizing information. Three different sizing options are available as part of the model hierarchy:-

- ❖ Object Points
- ❖ Function Points
- ❖ Lines of Source Code

The COCOMO II application composition model uses object points.

Like function point, the object point is an indirect software measure that is computed using counts of the number of.

- ❖ **Screens** (at the user interface),
- ❖ **Reports.**

COST ESTIMATION FOR THIS PROJECT

(1) SCREENS

- i. Home Page
- ii. Registration for Patient.
- iii. Patient update details
- iv. Book Appointment
- v. View Appointment Status
- vi. Cancel Appointment

(2) REPORTS

- i. Total Visitors on Website.
- ii. Total Patients Treated.
- iii. Total Appointments Taken.
- iv. Total Appointments Cancelled.
- v. Total Doctors on Leave.
- vi. Total Doctors Added.
- vii. Total Doctors Removed.
- viii. Total Consultant Fee Collected.

CHAPTER 5

RISK ANALYSIS

Risk analysis is the process of identifying and analyzing potential issues that could negatively impact key business initiatives or projects. This process is done to help organizations avoid or mitigate those risks.

Performing a risk analysis includes considering the possibility of adverse events caused by either natural processes, such as severe storms, earthquakes or floods, or adverse events caused by malicious or inadvertent human activities. An important part of risk analysis is identifying the potential for harm from these events, as well as the likelihood of their occurrence.

S.No	RISK	CATEGORY	PROBABLITY (P)	IMPACT (I)	RMMM PLAN
1.	SOME TEAM MEMBER BECOME SICK IN BETWEEN	TECHNICAL RISK	20%	2	OTHER TEAM MEMBERS DISTRIBUTE THE WORK IN BETWEEN THEM
2.	DELIVERY DEADLINE TIGHTENED	PROJECT RISK	30%	1	TEAM MAY USE EXTRA MEMBERS TO DO JOB ON SCHEDULED TIME
3.	LOSING OF ALL PROJECT DATA THIS MAY HAPPEN DUE TO HARD DISK FAILURE	PROJECT RISK	20%	2	BACK UP THE PROJECT ONLINE OR IN EVERY SYSTEM OF EVERY MEMBER
4.	TEAM DISTENTION / LACK OF COHESION	PROJECT RISK	10%	3	WE MAKE SOME RULES HOW WE CONSULT EACH OTHER

CHAPTER 6

TESTING

Testing is a critical phase in the development of the Hospital Management System (HMS) website. It ensures that the system functions correctly, meets the specified requirements, and provides a smooth user experience. Below is a detailed testing strategy that includes various testing types, test cases, and approaches.

6.1 Types of Testing

- ❖ Unit Testing
- ❖ Objective: To verify that individual components or modules function correctly.
- ❖ Scope: Test individual functions, methods, and classes.
- ❖ Tools: JUnit (for Java), NUnit (for .NET), PyTest (for Python).

6.1.2 Integration Testing

- ❖ Objective: To ensure that integrated modules work together as expected.
- ❖ Scope: Test interactions between integrated modules.
- ❖ Tools: Selenium, JUnit, Postman (for API testing).

6.1.3 System Testing

- ❖ Objective: To validate the entire system's functionality against requirements.
- ❖ Scope: Test the complete system, including all modules and interactions.
- ❖ Tools: Selenium, QTP, Test Complete.

6.1.4 User Acceptance Testing (UAT)

- ❖ Objective: To ensure the system meets user requirements and is ready for deployment.
- ❖ Scope: Conducted by end-users in a real-world environment.
- ❖ Tools: Manual testing based on user scenarios.

6.1.5 Performance Testing

- ❖ Objective: To evaluate the system's performance under various conditions.
- ❖ Scope: Load testing, stress testing, and scalability testing.
- ❖ Tools: JMeter, Load Runner.

6.1.6 Security Testing

- ❖ Objective: To identify vulnerabilities and ensure data protection.
- ❖ Scope: Test for SQL injection, XSS, CSRF, and other security threats.
- ❖ Tools: OWASP ZAP, Burp Suite.

6.2 Maintenance and support

You need to perform regular website maintenance. Most website maintenance can be self-service but it's better to hire a website maintenance service to help you with it. This way, you can focus on growing your business and worry less about maintaining your website.

Your hospital's website is a vital tool for patients to find information, book appointments, and access services. If it's not regularly maintained, it can lead to broken links, outdated information, and a poor user experience. With many healthcare providers going digital, it's important to ensure your website ranks highly on search engines and stands out from the competition. By investing in regular website maintenance, you can focus on providing excellent patient care while professionals keep your website in top shape.

6.2.1 Updating Website Software

If you are using any kind of content management system like WordPress or any kind of script like PHP, it needs to always be up to date. WordPress usually updates itself automatically but there are instances where it might need to be updated manually.

Additionally, if you are using a website host, a web service maintenance provider will ensure that they are updating their core server software on a regular basis. For instance, they will check if things like FTP service running on the server is up to date. If however, you run your own web server, they will manually test and apply updates.

6.2.2. Improving Website Speed

Did you know that twenty five percent of users navigate out of a website if it takes more than three seconds to load? Page loading is obviously an important part of the user experience. However, we tend to let it slide in order to accommodate new nifty functionality or esthetic website design.

In reality, web visitors do not care about all the bells and whistles. They care about accessing the information they want as fast as possible. Another important factor to consider is how page load speed affects your search engine rankings. A slow website increases bounce rate and if people are navigating out of your web page as fast as they got there, then Google will think your website isn't important so they will push it further down the rankings.

A website support service will always check your website's speed by constantly using the insights that they get from speed-checking tools to continually improve how it loads.

6.2.3. Fixing HTML Errors

HTML Code isn't easy especially if you know nothing about website coding. You need to pay attention to detail to ensure that it's working well. What might seem like a minor error could lead to a number of problems such as funky looking pages, missing

multimedia, etc. Reliable website support should be able to inspect, find and fix issues related to HTML.

6.2.4. Backing up Files

Imagine how frustrating it would be if you were to lose all your website's essential files. This includes images, pages, blog posts, plugins, etc. If you run your website on WordPress, you might have a few automated backup options. Website maintenance services ensure that file and database backups are automatically performed at least on a weekly basis.

6.2.5. Developing New Content

Your content strategy might be on point but it's always a good idea to improve it by creating new types of content. Your web content also needs to be up to date. If you wrote a post that was published years ago, it might not be relevant right now and some of the information might need to be updated.

6.2.6. Search Engine Optimization

Regularly checking your website for areas where searchability can be improved can go a long way to improving search engine optimization. Your blog posts can be altered to include popular and competitive keywords. Your landing pages can also be updated to include more relevant and timely keywords. Even your "About Us" page can be updated regularly to take advantage of popular keywords in your industry. Keyword research is also important when creating new content for your website.

As part of the SEO strategy, it's also important to identify structural issues within your website that may affect how search engines view your site.

6.2.7. Ensuring Design Consistency across All Pages

Your website should function properly in all the latest versions of major website browsers like Chrome, Safari, Firefox, etc. These browsers update frequently and if your website doesn't adapt to the updates it might not show up or it might not look the way you want it to. This could negatively affect your business and your branding. It's also important to ensure that you have a responsive website design that looks good on mobile devices.

Another factor to consider about your website's design is uniformity. If you have a lot of branding elements that represent your company, they should be included uniformly across all your pages.

6.2.8. Fixing Broken Links

Every link on your website should lead exactly to the right place but sometimes links can go bad. A site or resource you linked to earlier might disappear, a page on your website might become unavailable or you might move a post and forget to update others that link to it.

Broken links can be detrimental to your website. They could give visitors the impression that:

- ❖ Your website is not user friendly
- ❖ You don't regularly update your website
- ❖ You are not credible

Ideally, a website maintenance service will routinely check your website to identify any broken links. As soon as they find them, they will either modify or remove them.

6.2.9. Reviewing Your Website Analytics

Your website's performance needs to be monitored to ensure that everything is working as it should. An experienced website support specialist will examine high-level metrics like how many new and returning visitors your website has received for the past week or which blog posts are the most read. They will then make adjustments your website accordingly to ensure all the metrics are at their peak.

This is just a minimum starting point of what website maintenance is all about. Some of these tasks can be overly time consuming. Additionally, if you do not know what you are doing, you could end up doing more damage. The better option is to hire professional website maintenance services.

6.3 Security

It's no secret that security is critically important for healthcare organizations. Patients need to know their data is safe. Having a site with technical problems not only puts you at risk of a data breach, but it also hurts your brand. Here are seven areas you should focus on to ensure your site is helping you deliver on your promises.

6.3.1Protect user data.

Healthcare organizations must follow HIPAA guidelines, meaning you must comply with requirements to protect the privacy and security of health information. Your website should use secure sockets layer (SSL) technology to securely encrypt necessary page, form, transactional data, and protected health information (PHI) from the web browser to the server. PHI should be encrypted in transit and at rest. That means when it's being transmitted between systems, such as data sent when a user submits a form, as well as when that data is stored for future use, often in a database. Online forms that accept user data should be managed using specific protocols to comply with HIPAA. Protecting the submitted data in forms is critical. Compliant content management systems (CMS) like Vital Site through form builders like Formulate capture and store specific information when your users view a form submission and creates an audit trail. It logs the user ID, date and time stamp, IP address, and location and state of the data at the time it was accessed.

If you accept online bill payment or online donations, you must also be payment card industry (PCI) compliant and ensure debit and credit card information remains secure throughout the transaction process. Once submitted, payment data should be transmitted immediately to the payment processor and never stored.

6.3.2 Protect administrative accounts.

As part of staying in compliance with HIPAA, and to prevent unauthorized changes on your website, administrative accounts in the CMS must be monitored and protected. With VitalSite, passwords are encrypted and stored using SQL Server encryption. All of this ensures that it's difficult for an attacker to access and use someone's password. In addition, other safeguards, such as requiring strong passwords and periodic passwords changes, as well as locking inactive accounts, are also considered industry best practices for account protection.

6.3.3 Stay up to date.

One of the primary ways hospital websites fall out of compliance is by falling behind with CMS upgrades, updates, and security patches to the software administrators don't see, including web servers, databases, programming languages and framework. Falling behind on updates and upgrades are common ways organizations that run on a platform like WordPress find themselves vulnerable with multiple plug-ins requiring their own updates and patches. Ensure you're on the latest version of your CMS and look to partner with an agency that provides upgrades as part of the licensing, as Geonetric does with VitalSite.

6.3.4 Review your network architecture.

Whether you manage web hosting yourself or use an agency or third party, ensuring there are tools and processes in place to minimize the risk of a breach is essential. Especially today, where cyber-attacks are all too common. Areas to look for include redundant web application firewalls, intrusion detection systems, and systems to protect from common web attacks like distributed denial of service (D/DoS) and SQL injection.

6.3.5 Ensure 24/7 uptime monitoring and support.

Your web partner should know before you do that your site is down and respond around the clock with a team who can fix the problem regardless of whether it's a server, network, software, or content issue.

6.3.6 Protect from downtime with scalability.

Reliable networks are more important than ever, which means having scalability in your network is essential – especially when you consider you never know when a surge in traffic will arise. Case in point: healthcare websites saw huge traffic surges as critical COVID-19 communications, testing, and vaccine announcements were released. These traffic spikes need to be considered before they happen. Without auto-scaling in place, availability can be disrupted from unexpected traffic surges or a failure in one part of the system, bringing down the entire site. With proper auto-scaling cloud infrastructure, your site will remain available without missing a beat.

6.3.7 Require backups.

If the worst happens, how much would you lose and how quickly can you be up and running again? These are the questions you should ask. At Geonetric, we perform a full backup of your entire site and database each day. Every 15 minutes we perform transaction log backups and save these for two days. In addition, daily backups are saved for a week; weekly backups are saved for four weeks; and monthly backups are saved for a year. We also perform database consistency checks nightly. It's also important to ensure there is proper redundancy for quick recovery purposes.

CHAPTER 7

CONCLUSION

I have successfully designed and developed webpage, which is demonstrated in the way so that all the patients can make their appointment after that staff will align it and make for n appointment patient according to doctor schedule. The system was implemented using webbased technologies which include CSS, JS, Bootstrap, HTML, MySQL and PHP. The project can be entered using a username and password. Where User can search various services available, Book appointment, view doctor details and Doctor can view their appointments and patient detail using their id. It is accessible either by an admin or staff. Only they can add data into the database. The data can be retrieved easily. The interface is very user-friendly Hospital Management System is Powerful, Flexible and easy to use and is designed and developed to deliver real conceivable benefits to hospitals, clinics etc.

The project was successfully completed after a lot of efforts and work hours. This project underwent number of compiling, debugging, removing errors, making it bug free, adding more facilities in Hospital Management System and interactivity making it more reliable and useful.

This project focused that scheduling a project and adhering to that schedule creates a hard sense of time- management. It has also let us known that co-operative teamwork always produce effective results.

The entire project has been developed and deployed as per the requirements stated by the user. It is found to be bug free as per the testing standards that are implemented.

Finally, we like to conclude that we put all our efforts throughout the development of our project and tried to fulfil most of the requirements of the user.

7.1 Reference

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